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A comparative study of algorithm for fluid simulation of high density plasma discharges

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Abstract.

A new algorithm for steady-state fluid simulation of high density, low temperature plasma discharge is suggested. The restriction on simulation time step and a new method to overcome it is discussed. For a comparison of the new method with a previous other method, a one-dimensional fluid simulation of inductively coupled plasma discharge is studied.

Keywords: plasma, material processing, fluid simulation

PACS: 52.25.Fi

INTRODUCTION

Low temperature, high density plasma sources are widely used for electronic device fabrications such as semiconductors, TFT-LCDs, and solar cells. The inductively coupled plasma or the capacitively coupled plasma reactors are typical ones in these material processes. For this reason, there were many research results in modeling and simulation for these sources[1, 2, 3, 4]. For transport modeling of high density discharge, fluid simulation can be adopted because the profiles of densities, temperatures, fluxes and potentials are easily obtained.

It is known that there is a strong restriction on the simulation time step(Δt), because the shielding time scale of an electric field perturbation is very short. The maximum time step is related to various time scales such as collision frequency, Courant time step, plasma frequency, and dielectric relaxation time. In an explicit scheme, the time step cannot exceed the minimum of those time scales. For the typical ICP discharges, the time step cannot be larger than the dielectric relaxation time, which is about 10^{-13} sec. There are several methods to overcome this restriction[5, 6, 7, 8].

In this work, a new method for steady-state fluid simulation of high density plasma discharge is suggested by modifying the explicit form of Poisson's equation. A comparison of the new method is given with the previously known one is given to study the difference and similarities.

DESCRIPTION OF THE NEW METHOD

The usual set of equations for fluid simulation of high-density plasma discharges are the continuity, momentum, electron energy, and Poisson equations.

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \vec{\Gamma}_e = SL_e \quad (1)$$

$$\frac{\partial n_i}{\partial t} + \nabla \cdot \vec{\Gamma}_i = SL_i \quad (2)$$

$$\frac{\partial \vec{\Gamma}_e}{\partial t} + \nabla \cdot (\vec{v}_e \vec{\Gamma}_e) = \frac{en_e}{m} \nabla V - v_{eN} \vec{\Gamma}_e - \frac{1}{m} \nabla (n_e T_e) \quad (3)$$

$$\frac{\partial \vec{\Gamma}_i}{\partial t} + \nabla \cdot (\vec{v}_i \vec{\Gamma}_i) = -\frac{Z_i en_i}{M} \nabla V - v_{iN} \vec{\Gamma}_i - \frac{1}{M} \nabla (n_i T_i) \quad (4)$$

$$\frac{\partial}{\partial t} \left(\frac{3}{2} n_e T_e \right) + \nabla \cdot \left(\frac{5}{2} \vec{\Gamma}_e T_e - \frac{5}{2} \frac{en_e T_e}{m v_{eN}} \nabla T_e \right) - e \nabla V \cdot \vec{\Gamma}_e = P_{abs} - P_{loss} \quad (5)$$

$$\nabla^2 V = -e \frac{Z_i n_i - n_e}{\epsilon_0} \quad (6)$$

where $n_{i/e}$ is ion/electron density, $\vec{\Gamma}_{i/e}$ is ion/electron flux, $SL_{i/e}$ is source & loss of ions/electrons, $v_{iN/eN}$ is ion/electron neutral collision frequency, Z_i is the charged number of ions, P_{abs} is absorbed power density, P_{loss} is power loss density, and V is electrostatic potential. Using the drift diffusion approximation or the effective electric field, the left hand side of the Eqs. (3, 4) can be ignored. Because charge density (ρ) is $Z_i n_i - n_e$, the time derivative of the charge density can be expressed as

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= \frac{\partial (Z_i n_i - n_e)}{\partial t} \\ &= e Z_i \frac{\partial n_i}{\partial t} - e \frac{\partial n_e}{\partial t} \\ &= e Z_i \left(-\nabla \cdot \vec{\Gamma}_i + SL_i \right) - e \left(-\nabla \cdot \vec{\Gamma}_e + SL_e \right) \\ &= -e \nabla \cdot \left\{ -\frac{e Z_i^2 n_i}{M v_{iN}} \nabla V - \frac{\nabla (n_i Z_i T_i)}{M v_{iN}} - \frac{en_e}{m v_{eN}} \nabla V + \frac{\nabla (n_e T_e)}{m v_{eN}} \right\} + e (Z_i SL_i - SL_e) \\ &= \nabla \cdot \left[\left(\frac{e^2 n_e}{m v_{eN}} + \frac{e^2 Z_i^2 n_i}{M v_{iN}} \right) \nabla V \right] - \nabla^2 \left(\frac{en_e T_e}{m v_{eN}} - \frac{e Z_i n_i T_i}{M v_{iN}} \right) + e (Z_i SL_i - SL_e) \\ &= \epsilon_0 \nabla \cdot \left(\frac{1}{\tau} \nabla V \right) - \nabla^2 \left(\frac{en_e T_e}{m v_{eN}} - \frac{e Z_i n_i T_i}{M v_{iN}} \right) + e (Z_i SL_i - SL_e) \end{aligned}$$

where $\frac{1}{\tau} = \frac{e^2 n_e}{m v_{eN}} + \frac{e^2 Z_i^2 n_i}{M v_{iN}}$ is the dielectric relaxation time.

The last equation can be rewritten as

$$\frac{\partial \rho}{\partial t} + \frac{\rho}{\tau} = \epsilon_0 \nabla \cdot \left(\frac{1}{\tau} \nabla V \right) - \left[\nabla^2 \left(\frac{en_e T_e}{m v_{eN}} - \frac{e Z_i n_i T_i}{M v_{iN}} \right) - e (Z_i SL_i - SL_e) \right] \quad (7)$$

By focusing on the change of charge density and potential during the simulation time step, and regarding the remaining terms as slowly varying quantities, charge density can be obtained from a direct integration as

$$\rho^{(A)}(t) \simeq \rho(0) e^{-\frac{t}{\tau}} + \left[\epsilon_0 \nabla \cdot \left(\frac{1}{\tau} \nabla V \right) - \left\{ \nabla^2 \left(\frac{en_e T_e}{m v_{eN}} - \frac{e Z_i n_i T_i}{M v_{iN}} \right) - e (Z_i SL_i - SL_e) \right\} \right] \tau \left(1 - e^{-\frac{t}{\tau}} \right) \quad (8)$$

Here $\rho^{(A)}$ means analytic solution.

Assume the charge density is composed of analytic and numerical parts as $\rho(t) = \alpha \rho^{(A)} + (1 - \alpha) \rho^{(N)}$. Direct integrations of $\rho^{(A)}$ and $\rho^{(N)}$ from t to $t + \Delta t$ can be expressed as

$$\begin{aligned} \int_t^{t+\Delta t} \rho^{(A)} &= \rho(t) \tau \left(1 - e^{-\frac{\Delta t}{\tau}} \right) - \left[\epsilon_0 \nabla \cdot \left(\frac{1}{\tau} \nabla V \right) - \left\{ \nabla^2 \left(\frac{en_e T_e}{m v_{eN}} - \frac{e Z_i n_i T_i}{M v_{iN}} \right) - e (Z_i SL_i - SL_e) \right\} \right] \tau^2 \left(1 - e^{-\frac{\Delta t}{\tau}} \right) \\ &+ \left[\epsilon_0 \nabla \cdot \left(\frac{1}{\tau} \nabla V \right) - \left\{ \nabla^2 \left(\frac{en_e T_e}{m v_{eN}} - \frac{e Z_i n_i T_i}{M v_{iN}} \right) - e (Z_i SL_i - SL_e) \right\} \right] \tau \Delta t \end{aligned} \quad (9)$$

$$\int_t^{t+\Delta t} \rho^{(N)} = \rho^{(N)}(t + \Delta t) \Delta t \quad (10)$$

By integrating Eq. (7) from t to $t + \Delta t$ and dropping superscripts (A) and (N), charge density can be expressed as

$$\begin{aligned} \rho(t + \Delta t) &= \rho(t) - \alpha \rho(t) \left(1 - e^{-\frac{\Delta t}{\tau}} \right) - (1 - \alpha) \rho(t + \Delta t) \frac{\Delta t}{\tau} \\ &+ \left[\epsilon_0 \nabla \cdot \left(\frac{1}{\tau} \nabla V \right) - \left\{ \nabla^2 \left(\frac{en_e T_e}{m v_{eN}} - \frac{e Z_i n_i T_i}{M v_{iN}} \right) - e (Z_i SL_i - SL_e) \right\} \right] \left[\alpha \tau \left(1 - e^{-\frac{\Delta t}{\tau}} \right) - \alpha \Delta t + \Delta t \right] \end{aligned} \quad (11)$$

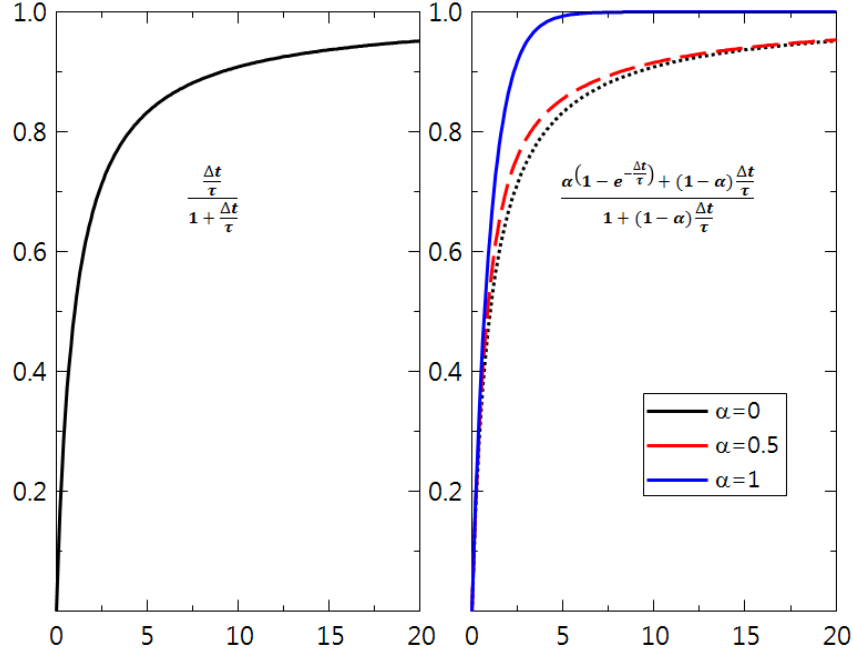


FIGURE 1. Comparison of functions in two methods

Using $\rho(t + \Delta t) = -\epsilon_0 \nabla^2 V(t + \Delta t)$, Poisson's equation can be written as

$$\begin{aligned} \nabla^2 V(t + \Delta t) &+ \nabla \left(\frac{1}{\tau} \right) \cdot \nabla V(t + \Delta t) \tau \frac{\alpha \left(1 - e^{-\frac{\Delta t}{\tau}} \right) + (1 - \alpha) \frac{\Delta t}{\tau}}{1 + (1 - \alpha) \frac{\Delta t}{\tau}} \\ &= -\frac{\rho(t)}{\epsilon_0} \frac{1 - \alpha \left(1 - e^{-\frac{\Delta t}{\tau}} \right)}{1 + (1 - \alpha) \frac{\Delta t}{\tau}} + \frac{1}{\epsilon_0} \left\{ \nabla^2 \left(\frac{en_e T_e}{mv_{eN}} - \frac{eZ_i n_i T_i}{Mv_{iN}} \right) - e(Z_i S L_i - S L_e) \right\} \tau \frac{\alpha \left(1 - e^{-\frac{\Delta t}{\tau}} \right) + (1 - \alpha) \frac{\Delta t}{\tau}}{1 + (1 - \alpha) \frac{\Delta t}{\tau}} \end{aligned} \quad (12)$$

RESULTS AND DISCUSSIONS

Equation (12) is derived from an analytic and semi-implicit part of the charge density. In Eq. (12), α is used as a weighting parameter. If $\alpha = 0$, which means the removal of the analytic part $\rho^{(A)}$, the equation can be written as

$$\begin{aligned} \nabla^2 V(t + \Delta t) &+ \nabla \left(\frac{1}{\tau} \right) \cdot \nabla V(t + \Delta t) \tau \frac{\frac{\Delta t}{\tau}}{1 + \frac{\Delta t}{\tau}} \\ &= -\frac{\rho(t)}{\epsilon_0} \frac{1}{1 + \frac{\Delta t}{\tau}} + \frac{1}{\epsilon_0} \left\{ \nabla^2 \left(\frac{en_e T_e}{mv_{eN}} - \frac{eZ_i n_i T_i}{Mv_{iN}} \right) - e(Z_i S L_i - S L_e) \right\} \tau \frac{\frac{\Delta t}{\tau}}{1 + \frac{\Delta t}{\tau}} \end{aligned} \quad (13)$$

This expression is the method in Ref. [6], which is numerically stable.

If $0 < \alpha < 1$, Eq. (12) becomes a similar but different form compared with Eq. (13). Basically, the difference comes from $\frac{\alpha \left(1 - e^{-\frac{\Delta t}{\tau}} \right) + (1 - \alpha) \frac{\Delta t}{\tau}}{1 + (1 - \alpha) \frac{\Delta t}{\tau}}$ in Eq.(12) and $\frac{\frac{\Delta t}{\tau}}{1 + \frac{\Delta t}{\tau}}$ in Eq.(13). In figure (1), functions are compared as a function of $\frac{\Delta t}{\tau}$. From this, it can be found that the new method gives a numerically stable solution similar to the method in Ref. [6]. Also the new method has an exponential decay form, which is similar to the method in Ref. [7]. Accordingly, it converges fast to the solution, and can be applied to the case when the local ambipolarity is not satisfied ($\vec{J}_{tot} \neq 0$).

CONCLUSIONS

We developed a new method to overcome the time step restriction in the steady-state simulation of high density plasma discharges based on the analytic method, and this was compared with a previously known method. The method is similar to methods in Ref. [6, 7]. Because of the exponential decay form in the expression of charge density, the present method is numerically stable, and it converges fast to the solution. For the larger reactors such as display device manufacturing whose dimension is about 3-4 m, this method with the scheme in mesh size [9] will be a useful one in plasma simulation.

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